Xuan Ouyang

xuan.d.ouyang@gmail.com | linkedin.com/in/xuanouyang/ | github.com/XuanOuyang

EDUCATION

University of Washington **Expected: June 2027**
Bachelor of Science in Data Visualization
Minor: Computer Science & Software Engineering, Data Science

- **Certifications**: Quantum Computing (Qubit x Qubit) | TIP102 (CodePath) | Claude 101 & AI Fluency (Anthropic)
- **Organizations**: CodePath, QxQ Quantum Computing, Game Heads, TrickFire Robotics, Rewrite the Code, Basta

WORK EXPERIENCE

AI & Community Technology Developer Intern, *Be the Blessing* **June 2026 – August 2026**

- Built the foundation for an **AI-assisted journaling app,** integrating the Claude API to analyze user-written journal entries, generate contextual responses, and surface relevant wellness resources from a content library.
- Engineered a scalable content architecture supporting **40+ wellness lessons and quests** across 3 age groups, enabling personalized goals and age-specific experiences.
- Developed **API request, response-handling, and input-processing logic** to integrate AI-generated recommendations into the mobile application workflow.
- Evaluated **AI APIs, cloud infrastructure, privacy requirements, and development frameworks** to determine technical feasibility and inform the MVP architecture.

Lead Game Developer, *Gameheads* **June 2026 - August 2026**

- Led a **3-person programming team**, managing the GitHub master branch and coordinating multiple development branches to integrate contributions and resolve merge conflicts throughout development.
- Optimized player and enemy behavior using the **State design pattern**, improving game performance from approximately **12 FPS to 60 FPS** during real world gameplay.
- Designed an extensible projectile system using **C# inheritance and polymorphism**, extending a base projectile class with specialized piercing behavior and a **HashSet** to prevent duplicate enemy collisions.
- Implemented enemy AI using **NavMesh pathfinding** and **ray-casting systems** for player point-of-view detection and shooting trajectory visualization, supporting core combat and gameplay mechanics.

Web-Development Pre-Apprentice, *Computing for All* **May 2024 - August 2024**

- Developed the Back-End functionality for a **responsive multi-page web application**, implementing username/password forms, password requirements, interactive components, animations, and user streak tracking across desktop, tablet, and mobile layouts.
- Structured application data using **JavaScript arrays and objects** to manage user information and streaks; debugged JavaScript logic and iteratively resolved functionality issues throughout development.
- Collaborated on a 2-person development team utilizing Git and GitHub, **managing feature branches** and integrating JavaScript changes with HTML/CSS components through a shared development workflow.

PROJECTS

House Plant Identifier | Python, Keras, TensorFlow

- Built and trained neural networks achieving **97% accuracy** across **47 plant species** using 15,000 labeled images
- **Optimized model performance by 14%** through hyperparameter tuning of batch size and training epochs
- Developed a GPU-accelerated training pipeline using TensorFlow, Keras, NumPy, Pandas, and OpenCV

Lynnwood School District Data Analytics Dashboard | Python, SQL, Tableau

- Analyzed district-wide survey data from **15,000+ students** to geographically map bus-dependent populations and identify underserved zones using Pandas, Tableau, and SQL
- Built **interactive Tableau dashboards** from processed datasets to visualize geographic trends and provide insights to stakeholders
- Identified high-need geographic clusters that directly informed district **leadership's evaluation** of school bus program expansion and route changes

Endangered Animal Website | HTML, CSS, JavaScript

- Implemented **dynamic DOM manipulation and event-driven interactions** with JavaScript
- Optimized **responsive layouts and client-side performance** across screen sizes

SKILLS & INTERESTS

Programming Languages: Python, SQL, Java, HTML, CSS, JavaScript, C#, C++ ,C
Tools/Frameworks: TensorFlow, Keras, Pandas, NumPy, OpenCV, Unity, Git/GitHub, Tableau, PostgreSQL, Claude API
Interests: Artificial Intelligence, Machine Learning, Automation, Quantum Computing